1. Structure of Real Datasets

The `Realdataset/` directory contains both the raw input data components along with their preprocessing pipelines, and the finalized preprocessed benchmark slices.

1.1 Raw Data & Preprocessing Pipeline

To ensure complete transparency, we provide the full data cleaning workflow starting directly from the unmanipulated raw matrices.

- **Pipeline Script**: `RealData/code/data_preprocess.R`
- **Functionality**: This script illustrates the comprehensive data cleaning, quality control (QC), spot filtering, and cell-type deconvolution matching implemented for all real-world slices.

The layout of the raw files inside `code_and_data/data/raw_data/` is structured as follows:

1.1.1 DLPFC Raw Dataset (`raw_data/dlpfc_dataset/`)

- `151507_filtered_feature_bc_matrix.h5` to `151676_filtered_feature_bc_matrix.h5`: 12 standardized HDF5 files containing raw spatial transcriptomic expression matrices.
  - Slices 1–4 (151507–151510): 4 consecutive tissue slices from **Donor 1**.
  - Slices 5–8 (151669–151672): 4 consecutive tissue slices from **Donor 2**.
  - Slices 9–12 (151673–151676): 4 consecutive tissue slices from **Donor 3**.
- `assays.h5`: Raw expression count matrix for the single-nucleus RNA-seq (snRNA-seq) reference.
- `se.rds`: Complete metadata and cellular annotation object for the snRNA-seq reference.
- `result_matrix.RData`: A pre-extracted and optimized copy of the snRNA-seq matrix for expedited execution.
- `position1.txt` to `position12.txt`: Individual spot spatial coordinates matching the 12 ST HDF5 matrices respectively (corresponding to the official `tissue_positions_list.txt` from the spatialLIBD repository).

Note: Due to GitHub's file size limitations, ultra-large raw baseline references (`assays.h5`) exceed the upload threshold and cannot be directly archived in this repository. Users **must** manually download the complete reference package via Dropbox Link (https://www.dropbox.com/s/5919zt00vm1ht8e/sce_DLPFC_annotated.zip?dl=1), unzip it, and place the extracted files directly into the `data/Realdataset/raw_data/dlpfc_dataset/` directory before executing the preprocessing pipeline.

1.1.2 SCC Raw Dataset (`raw_data/scc_dataset/`)

- `scc_P9_rep1.tsv` to `scc_P9_rep3.tsv`: Spatial transcriptomics count matrices (in TSV format) corresponding to 3 technical replicates sampled from **Patient 9**.
- `spot_P9_rep1.tsv` to `spot_P9_rep3.tsv`: Spatial coordinate metadata matrices (in TSV format) mapping spots to physical locations for Patient 9's replicates.
- `merge10pts_counts.txt`: Raw count matrix of the single-cell RNA-seq (scRNA-seq) benchmark reference encompassing ten integrated patients.
- `GSE144236_patient_metadata_new.txt`: Clinical and cell-type annotation dictionary for the scRNA-seq reference dataset.
- `scc_sc_count.RData`: Pre-processed and serialized scRNA-seq reference matrix ready for rapid memory allocation.

Workflow Option: Reproduction users can fully regenerate the benchmarking input profiles from scratch by executing `data_preprocess.R` over these raw directories, or directly leverage the pre-computed matrices detailed in Section 1.2.

1.2 Structure of Preprocessed Real Datasets

The preprocessed real datasets are stored in the `Realdataset/` directory, which contains three subdirectories:

- `dlpfc_acrossdonor/`
- `dlpfc_samedonor/`
- `scc/`

Each subdirectory contains a collection of CSV files corresponding to individual tissue slices.

1.2.1 File Naming Convention All files follow the naming pattern: matrix{sliceID}_{datatype}_{suffix}.csv

where:

- `{sliceID}` denotes the slice index
  - For DLPFC datasets: values range from 1 to 4
  - For SCC dataset: values range from 1 to 3
- `{datatype}` indicates the data type:
  - `count` — Gene expression count matrix (gene × spot)
  - `position` — Spatial coordinate matrix (spot × location)
  - `celltype` — Covariate matrix (spot × covariate)
- `{suffix}` identifies the dataset:
  - `acrossdonor` — DLPFC across-donor integration
  - `samedonor` — DLPFC same-donor integration
  - `scc` — SCC dataset

1.2.2 Data Structure For each slice, the following matrices are provided:

- **Count matrix (`count`)**
  A gene × spot matrix containing UMI counts.
- **Position matrix (`position`)**
  A spot × location matrix containing spatial coordinates.
- **Cell-type matrix (`celltype`)**
  A spot × covariate matrix representing estimated cell-type proportions (used as covariates in the model).

2. Structure of Preprocessed Simulation data

The `Simulation data/` directory contains three main subdirectories:

- `basic simulation/`
- `basic simulation without covariates/`
- `additional simulation/`

Each folder stores pre-generated simulation datasets in `.RData` or `.csv` format.

(1) Basic Simulation (with Covariates)

This folder contains three subdirectories corresponding to the three canonical spatial patterns:

- `linear/`
- `focal/`
- `period/`

Each subdirectory includes files named as:

- data_{pattern}_seed{n}.RData
- data_{pattern}_seed{n}_paste.RData

where:

- `{pattern}` {linear, focal, period}
- `{n}` denotes the random seed index
- `_paste` indicates datasets integrated using the PASTE method

(a) Files without `_paste` For files such as:

- data_focal_seed1.RData

After loading, the object `all_results_list` is obtained. It contains three elements:

- `inf1` — low zero-inflation
- `inf3` — medium zero-inflation
- `inf5` — high zero-inflation

Each zero-inflation list contains:

- `data_list`: a list of 8 count matrices (spot $\times$ gene) corresponding to eight signal strengths (high, medium, medium, medium, low, low, very low, extremely low), covering the four integration Settings 1–4.

  -These eight signal levels are used to construct four integration settings:

  - **Setting 1**: element {1, 2, 3, 4}
  - **Setting 2**: elements {1, 2, 5, 6}
  - **Setting 3**: elements {3, 4, 5, 6}
  - **Setting 4**: elements {2, 5, 7, 8}

- `calpha_list`: a list of 8 covariate matrices (spot $\times$ covariate) corresponding to each signal-strength scenario.

(b) Files with `_paste` For files such as:

- data_focal_seed1_paste.RData

After loading, the object `data_paste_list` is obtained. It contains three elements corresponding to:

- `inf1` — low zero-inflation
- `inf3` — medium zero-inflation
- `inf5` — high zero-inflation

Each element contains four integrated count matrices (spot × gene), corresponding to the four integration Settings 1–4 (in order).

(2) Basic Simulation Without Covariates

This folder has the same structure as the basic simulation folder, with subdirectories:

- `linear/`
- `focal/`
- `period/`

File naming follows:

- data_{pattern}_nocov_seed{n}.RData
- data_{pattern}_nocov_seed{n}_paste.RData

(a) Files without `_paste` For example:

- data_focal_nocov_seed1.RData

After loading, `all_results_list` is obtained. It contains only one element:

- `inf3` (medium zero-inflation)

Inside `inf3`:

- `data_list`: contains 8 count matrices (spot × gene) corresponding to eight signal strengths under the four integration Settings 1–4.
- No covariate matrices are included.

(b) PASTE-integrated files For example:

- data_focal_nocov_seed1_paste.RData

After loading, `data_paste_list` is obtained. Only the second element (corresponding to `inf3`) contains values. It includes four integrated count matrices (spot × gene), corresponding to the four integration Settings.

(3) Additional Simulation

This folder contains nine additional heterogeneous spatial patterns:

- `Focal-Period`
- `Linear-Focal`
- `Linear-Period`
- `Polynomial1`
- `Polynomial2`
- `Polynomial3`
- `Polynomial4`
- `Sigmoid`
- `ZINNGP`

(a) Files without `_paste` Files are named as:

- dif_data_seed{n}{patternX}{patternY}.RData

where:

- `{n}` is the seed index
- `{patternX}` is the spatial pattern modeled along the x-axis
- `{patternY}` is the spatial pattern modeled along the y-axis

For example: dif_data_seed1polynomial1polynomial1.RData

After loading, the object `result_list` is obtained. It contains four lists, each corresponding to one slice. Each slice includes three matrices:

1. Count matrix (spot × gene)
2. Spatial coordinate matrix (spot × location)
3. Covariate matrix (spot × covariate)

(b) PASTE-integrated datasets Files such as: data_paste_polynomial1polynomial1_seed8.csv.

- represent PASTE-integrated count matrices (spot × gene).

3.Spatial Coordinate Generation

We used a fixed 32 × 32 grid for all simulations:

```
loc <- expand.grid(y = 0:31, x = 0:31)[, 2:1]
rownames(loc) <- paste("spot", 1:1024)
colnames(loc) <- c("x", "y")
```

data is saved in the `data/Simulation data/location.csv`.

Important Note on Covariates of PASTE-integrated datasets

For all simulation datasets, the PASTE method integrates only count matrices and does not integrate covariate information.Therefore, for integrated datasets, the covariate matrix from the first slice is used as the covariate.

Repository version information

The reproducibility package corresponds to GitHub commit:4353b68b3a00052d79ccbc88cbb7ef6d75f0b691. This commit represents the exact version used for the revised reproducibility package and supplementary materials.

Repository: https://github.com/zhoumeng123456/IBaySVG-analysis